CSA47 – DEEP LEARNING
CO1 – ASSIGNMENT 1

Analyse the Mathematical and Machine Learning Foundations of a Crop-Disease Screening System

K.Deeksha Sree(192424097)

CSA47 – Deep Learning
CO1 – ASSIGNMENT-1

CO1: Mathematical foundations of deep learning – linear algebra, probability and information theory, numerical computation, machine learning basics, capacity, overfitting and underfitting, estimation, bias and variance, Bayesian statistics, stochastic gradient descent, challenges motivating deep learning.

Assessment Title: Analyse the Mathematical and Machine Learning Foundations of a Crop-Disease Screening System (Total Marks - 50)

Nature of Assessment: Individual Duration: 1 Week Bloom's Level: K4 – Analyse
Submission Date: _____

Scenario — "AgriVision" Crop-Disease Screening

AgriVision is an agri-tech start-up that helps farmers detect leaf diseases from smartphone photographs. Before the company invests in a deep neural network, its Chief Data Scientist wants a rigorous analysis of the mathematical and machine learning foundations of the problem.

You will work with a public leaf-disease image dataset of your choice (for example, the tomato subset of PlantVillage, with Healthy, Early Blight and Late Blight classes). Images may be resized to 64 × 64 grayscale, giving 4,096 raw pixel features per image.

The dataset may be used at increasing scales:

$1{,}000 \rightarrow 5{,}000 \rightarrow 10{,}000 \rightarrow$ full dataset images

You are required to examine how the data is represented, break down the sources of error and uncertainty in classical machine learning models, differentiate their optimisation and generalisation behaviour, and determine, with evidence, whether deep learning is justified. Every preprocessing step must be documented and examined for its effect on the results.

Design Challenge:

AgriVision now needs to support 38 crop–disease classes, add images whose labels are about 20% noisy, and run the classifier on a low-cost smartphone within 50 ms and 5 MB.

Analyse how these changes affect your findings and determine whether your conclusion should be retained, modified or redesigned.

TASK 1 – Analyse the Problem Structure and Data Representation [K4] 4 Marks

Break down AgriVision into its learning-task components and examine how the choice of data representation influences the problem.

Discuss:

- The learning task, input, output and performance measure, and how each constrains the solution

- How the dataset, one image, a mini-batch of 32 images and the labels are organised as scalars, vectors, matrices or tensors, and why each shape matters
- How the L1, L2 and max (L∞) norms of the same feature vector differ, and which norm suits which analysis purpose
- How each preprocessing step (resizing, normalisation, flattening, data split) changes the information available to the model

You are expected to examine the consequences of each representation choice, not merely list the shapes.

Evidence Expected

A decomposition of the task into its components and an analysis of how representation and preprocessing affect the problem.

TASK 2 – Examine the Structure of the Feature Space using Linear Algebra [K4] 5 Marks

Use eigendecomposition and/or singular value decomposition to examine how information is distributed across the 4,096-dimensional image representation.

Discuss:

- How the eigenvalues or singular values are distributed, and what this reveals about redundancy among pixels
- The relationship between eigenvalues, singular values and explained variance
- How the reconstruction error changes as the number of components k varies, and where the trade-off point lies
- What the leading principal components represent when visualised as images
- How the cost of decomposition grows with the dataset size

Show at least one decomposition worked by hand on a small 2 × 2 or 3 × 3 sub-problem to support your analysis.

Evidence Expected

An analysis of the spectrum, an interpretation of the leading components and a justified compression trade-off point.

TASK 3 – Analyse Uncertainty using Probability and Information Theory [K4] 4 Marks

Examine the uncertainty in AgriVision's labels and predictions using probability and information theory.

Discuss:

- How the class imbalance is reflected in the entropy of the class distribution
- How the cross-entropy and KL divergence between the true and predicted distributions differ across your models, and what the differences reveal
- Why KL divergence is asymmetric, illustrated with your own numbers, and what each direction measures
- Why minimising cross-entropy and minimising KL divergence lead to the same classifier

You are expected to relate every quantity to the behaviour of the classifier, not merely compute it.

Evidence Expected

A comparative analysis of entropy, cross-entropy and KL divergence that explains model behaviour.

TASK 4 – Diagnose Numerical Computation Issues [K4] 3 Marks

Investigate where and why numerical failures occur when computing class probabilities and gradients.

Discuss:

- The conditions under which a naive softmax overflows or underflows (for example z = [1000, 999, 995])
- Why subtracting max(z) and the log-sum-exp trick remove the failure without changing the result

- How the condition number of the feature covariance matrix explains slow or unstable gradient-based training

Evidence Expected

A demonstrated failure traced to its cause, with an analysis of why the correction works.

TASK 5 – Differentiate Baseline Learning Strategies [K4] 4 Marks

Implement at least three meaningfully different learning algorithms on raw and PCA-compressed features and differentiate their behaviour.

The approaches must represent distinct learning strategies (for example: a linear parametric model, a non-parametric model and a kernel or tree-based model). Two settings of the same algorithm will not be considered as two different approaches.

Discuss:

- How the hypothesis space and inductive bias of each algorithm differ
- Which characteristics of 4,096-dimensional image data favour or hinder each algorithm
- How compression affects each algorithm differently, and why

Evidence Expected

Three distinct baselines with a reasoned analysis of why their performance differs.

TASK 6 – Analyse Capacity, Overfitting and Underfitting [K4] 4 Marks

Examine how model capacity drives training and validation error for your baselines.

Discuss:

- How varying capacity (polynomial degree, k in k-NN, tree depth, regularisation strength) changes training and validation error

- Where the underfitting and overfitting regions lie, and what causes each
- How the generalisation gap changes with capacity
- Why selecting hyperparameters on the test set biases the reported error, compared with a validation set or k-fold cross-validation

Do not simply present curves. You are expected to explain the causes behind each region.

Evidence Expected

Capacity–error curves with each regime identified and its cause explained.

TASK 7 – Decompose Error into Bias and Variance [K4] 4 Marks

Break down the error of your models into its components and examine the effect of prior knowledge on estimation.

Discuss:

- A bootstrap experiment (at least 10 replicas) separating bias2 and variance for a low- and a high-capacity model

- How closely the measured error matches bias2 + variance + noise, and why any gap exists
- How the MLE and MAP estimates of a class proportion differ under a Beta prior of your choice
- How the influence of the prior changes as the sample size grows

Evidence Expected

An experimental bias–variance decomposition and a comparative analysis of MLE and MAP estimates.

TASK 8 – Analyse the Behaviour of Gradient-Based Optimisation [K4] 4 Marks

Implement softmax regression from scratch and analyse how different optimisation strategies behave on it.

Discuss:

- The structure of the loss gradient and the terms that drive each update
- How batch gradient descent, stochastic gradient descent and mini-batch SGD differ in convergence path, noise and cost per epoch
- How at least three learning rates change stability and speed, and why
- How the observed behaviour relates to the conditioning identified in Task 4

You must show the derivation of the gradient to support your analysis.

Evidence Expected

A derived gradient and loss curves with an analysis of the differences between strategies and learning rates.

TASK 9 – Organise and Examine Experimental Performance [K4] 3 Marks

Organise the experimental results for all models across the dataset scales 1,000 → 5,000 → 10,000 → full, and examine the patterns they reveal.

Examine measures such as:

- Accuracy, precision, recall, F1-score and the confusion matrix
- Which classes are confused with each other, and why
- Training time, inference time, parameter count and memory footprint
- How each measure changes with dataset size and with compression

Use tables, graphs and performance plots that make the patterns and trade-offs visible.

Evidence Expected

Systematically organised results with the key patterns and trade-offs identified.

TASK 10 – Compare Theoretical Expectations with Experimental Behaviour [K4] 3 Marks

Examine the agreement between your mathematical analysis and the observed behaviour.

Discuss:

- Whether explained variance accounts for the accuracy retained after compression
- Whether optimisation speed agrees with the condition number and learning-rate analysis
- Whether the bias–variance results agree with the capacity curves
- The causes of any disagreement

Evidence Expected

A reasoned analysis of where theory and experiment agree or differ, and why.

TASK 11 – Analyse whether Deep Learning is Justified [K4] 4 Marks

Using your findings, analyse whether the limitations of classical machine learning on AgriVision's data warrant a deep network.

Discuss:

- Which limitations of the best baseline remain after tuning
- How the curse of dimensionality, local constancy and smoothness, and manifold structure manifest in your data
- What evidence indicates the need for a learned hierarchical representation
- Trade-offs in data, compute and interpretability

Your conclusion must be traced to evidence from your own analysis. Stating that deep learning is more accurate is not sufficient.

Evidence Expected

An evidence-based analysis of the need for deep learning, linked to specific findings.

TASK 12 – Analyse the Impact of Changed Requirements 8 Marks

Changed Requirement

AgriVision now needs to classify 38 crop–disease classes. Newly collected images have about 20% noisy labels, and the model must run on a low-cost smartphone within 50 ms per image and 5 MB of memory.

Re-evaluate your original design in response to these changed operating conditions. You may RETAIN → MODIFY → or REDESIGN your original solution.

A. Impact Analysis – 2 Marks

Discuss:

- Which elements of the original problem have changed
- How the larger label space affects entropy, cross-entropy loss and baseline accuracy
- How label noise affects bias, variance and the estimated error
- Which original assumptions about data and compute no longer hold

Evidence Expected

A clear breakdown of how each change affects the original analysis.

B. Examination of the Original Conclusion – 2 Marks

Discuss:

- Whether your best model fits within the 50 ms and 5 MB limits, using your measurements
- Which parts of its behaviour are most sensitive to 38 classes and noisy labels
- Which alternative approach is less affected, and why

Evidence Expected

A measured examination of the original conclusion under the new conditions.

C. Analysis of Adaptation Options – 2 Marks

Examine at least two options for responding to the changes, such as:

- Changing the PCA dimension or regularisation strength
- Using a noise-robust loss or label smoothing
- Compressing or simplifying the model

For each option, analyse its effect on capacity, error and computational cost.

Evidence Expected

A comparative analysis of adaptation options with their effects on capacity, error and cost.

D. Evidence-Based Conclusion – 2 Marks

State whether you retain, modify or redesign your conclusion, and trace it to:

- Mathematical analysis (entropy, bias–variance, conditioning)
- Experimental results
- Inference-time and memory measurements
- The trade-offs identified in Part C

Your conclusion must follow logically from your analysis.

Evidence Expected

A conclusion whose every element is traced to the analysis in Parts A–C.

Task 1 – Analyse the Problem Structure and Data Representation

1.1 Learning-task components

Component	AgriVision definition	Consequence for design
Task	Supervised, single-label, three-class image classification.	Learn a mapping from image pixels to a disease category.
Input	Smartphone leaf photograph; baseline representation is 64 × 64 grayscale.	Resizing and grayscale conversion reduce cost but can discard colour and fine lesion detail.
Output	One of Healthy, Early Blight or Late Blight; model may output three probabilities.	Use softmax probabilities and define an explicit decision rule.
Training signal	Image-label pairs; labels may contain mistakes in the changed scenario.	Label quality and class balance affect the learned decision boundaries.
Primary metric	Macro-F1 plus accuracy; per-class precision/recall and confusion matrix.	Macro-F1 gives each class equal weight and exposes poor minority-class performance.
Operational constraints	Fast, reliable screening; later 50 ms and 5 MB constraints.	A high-scoring but oversized or slow model is not deployable.

The goal is not merely to maximise training accuracy. A screening model should generalise to new farms, devices, lighting conditions and cultivars. If images from the same leaf or capture session leak across splits, reported performance will be optimistic.

1.2 Data structures and shapes

Object	Shape / type	Why it matters
One grayscale image	64 × 64 matrix (or tensor 64 × 64 × 1)	Rows and columns preserve spatial neighbourhoods; adjacent pixels often form edges and lesions.
Flattened image	Vector x ∈ R^4096	Compatible with classical vector models, but loses explicit spatial layout unless the model reconstructs it from feature positions.
Dataset of N images	X ∈ R^(N × 4096)	Rows are examples and columns are pixel features; scaling and memory costs grow with N × 4096.
Mini-batch of 32	X_b ∈ R^(32 × 4096)	Enables vectorised matrix operations and gives a gradient estimate from 32 examples.
Class labels	Integer vector y ∈ {0,1,2}^N; one-hot Y ∈ {0,1}^(N × 3)	Integer labels are compact; one-hot labels make cross-entropy and softmax-gradient formulas convenient.
Model scores	Z ∈ R^(32 × 3) for a batch	One score per example and class; softmax converts each row to probabilities.

1.3 L1, L2 and L-infinity norms

For x = [3, −4, 12], the L1 norm is $||x||_1 = |3| + |-4| + |12| = 19$; the L2 norm is $||x||_2 = \sqrt{(9 + 16 + 144)} = 13$; and the max norm is $||x||_\infty = \max(3,4,12) = 12$. L1 measures total absolute magnitude and often supports sparse regularisation; L2 measures Euclidean magnitude and is common in weight decay and reconstruction error; L^∞ captures the largest coordinate and is useful for worst-coordinate bounds. They answer different questions and should not be treated as interchangeable.

1.4 Preprocessing and information effects

- Resizing to 64 × 64: lowers memory and computation, but interpolation can erase tiny spots, lesion boundaries or texture. The interpolation method (area, bilinear, bicubic) must be recorded.
- Grayscale conversion: reduces three colour channels to one and cuts input dimensionality by roughly three. It can discard yellowing, chlorosis and colour contrasts that distinguish diseases. Compare grayscale with RGB if colour is diagnostically important.
- Normalisation: mapping integer pixels 0–255 to [0,1] improves numerical scale and optimisation. Standardisation using training-set means and standard deviations can further balance features, but must be fitted only on training data.
- Flattening: makes classical algorithms straightforward but discards image topology as an explicit inductive bias. A CNN retains local structure through convolutions.
- Data split: a test set estimates generalisation only if held out until the end. Random image-level splits can leak near duplicates; group-aware splitting is safer.

Conclusion for Task 1: representation is a modelling decision. The 4,096-vector is convenient for classical baselines, while the original image tensor is more suitable for models that exploit local spatial patterns.

Task 2 – Examine the Feature Space using Linear Algebra

2.1 PCA/SVD procedure

Let X be the N × 4,096 training matrix. Subtract each training feature mean to obtain X_c. The sample covariance is C = $X_c^T X_c/(N-1)$, a 4,096 × 4,096 matrix. PCA eigenvectors satisfy $C v_i = \lambda_i v_i$, with $\lambda_i \geq 0$. Alternatively, $X_c = U\Sigma V^T$; the right singular vectors (columns of V) are the principal directions and $\lambda_i = \sigma_i^2/(N-1)$. Explained-variance ratio for component i is $\lambda_i / \Sigma_j \lambda_j$, and cumulative explained variance through k is $\Sigma_(i=1)^k \lambda_i / \Sigma_j \lambda_j$.

Adjacent pixels are correlated because leaf surfaces, lighting gradients and lesions occupy connected regions. Consequently, the covariance spectrum is often concentrated: a relatively small number of components may explain broad illumination, leaf shape and large texture patterns, while many later components capture fine detail and noise. This is a hypothesis to test on the chosen split, not a universal numerical fact.

2.2 Hand-worked 2 × 2 eigendecomposition

Consider the illustrative covariance matrix C = [[2, 1], [1, 2]]. Its characteristic equation is $\det(C - \lambda I) = (2-\lambda)^2 - 1 = 0$, so $\lambda_1 = 3$ and $\lambda_2 = 1$. For $\lambda_1 = 3$, $(C-3I)v = 0$ gives $v_1 \propto [1,1]$, normalised to $[1/\sqrt{2}, 1/\sqrt{2}]$. For $\lambda_2 = 1$, $v_2 \propto [1,-1]$, normalised to $[1/\sqrt{2}, -1/\sqrt{2}]$. Total variance is 3+1=4; the first component explains 3/4 = 75%, and the second explains 25%. Keeping k=1 discards 25% of the variance in this toy example.

2.3 Reconstruction error and selecting k

For a centred row vector x, the rank-k reconstruction is $\hat{x}_k = \Sigma_(i=1)^k (x \cdot v_i) v_i$. The average squared reconstruction error equals the sum of discarded eigenvalues, $\Sigma_(i=k+1)^d \lambda_i$, when PCA is fitted with the standard covariance convention. It decreases monotonically as k grows, but classification accuracy need not: low-variance details can be disease-specific, and high-variance lighting or background effects can be irrelevant.

Illustrative k	Cumulative explained variance	Interpretation
32	70%	Strong compression; substantial information discarded.
64	82%	Potential efficiency point, but verify class-wise recall.
128	91%	More detail retained at moderate feature cost.

256	97%	Lower reconstruction error; diminishing compression benefit.

The values above are a placeholder example, not a measured PlantVillage spectrum. In the actual report, plot cumulative explained variance and validation macro-F1 against k. Select the smallest k whose validation performance is within a predeclared tolerance (for example, one percentage point of the best macro-F1), while satisfying latency and memory constraints.

2.4 Visual meaning of components and computation

Reshape each 4,096-element principal direction into a 64 × 64 image. Leading components often resemble smooth brightness gradients, broad leaf silhouettes or large-scale texture; later components tend to show finer edges and local patterns. Component images are directions of variation, not disease templates: a component may represent lighting or background rather than a causal symptom.

- Full covariance eigendecomposition requires $O(d^3)$ time and $O(d^2)$ memory, where d=4,096; storing a dense float32 covariance matrix alone requires about 64 MiB ($4{,}096^2 \times 4$ bytes).
- Forming X^TX costs approximately $O(Nd^2)$, so this becomes expensive as N grows. SVD of the data matrix can cost $O(Nd \cdot \min(N,d))$ in a conventional dense implementation; randomized/truncated SVD is often preferable when only a small k is needed.
- When N is much smaller than d, economy SVD or the N × N Gram matrix can be cheaper. The exact cost depends on algorithm and hardware.

Conclusion for Task 2: PCA is justified as a diagnostic and possible compression method, but k must be selected by validation classification performance and resource measurements, not explained variance alone.

Task 3 – Analyse Uncertainty using Probability and Information Theory

3.1 Class entropy and imbalance

For class proportions p_c, entropy is $H(Y) = -\Sigma_c\, p_c \log_2 p_c$. With three equally likely classes, $H(Y)=\log_2 3 \approx 1.585$ bits. If proportions are [0.70, 0.20, 0.10], then $H(Y)=-[0.70\log_2 0.70 + 0.20\log_2 0.20 + 0.10\log_2 0.10] \approx 1.157$ bits. Lower entropy indicates a more imbalanced label distribution. A majority-class classifier would obtain 70% accuracy in this example while having zero recall for two classes, illustrating why macro-F1 and class-wise recall are necessary.

3.2 Cross-entropy and KL divergence

For true distribution p and predicted distribution q, cross-entropy is $H(p,q)=-\Sigma\, p_c \log q_c$. KL divergence is $D_{KL}(p||q)=\Sigma\, p_c \log(p_c/q_c)$, and the identity $H(p,q)=H(p)+D_{KL}(p||q)$ holds when the same logarithm base is used. Since H(p) is constant for a fixed dataset and target distribution, minimising cross-entropy over model parameters also minimises KL divergence.

For a single true class with p=[1,0,0], suppose Model A predicts $q_A=[0.80,0.10,0.10]$ and Model B predicts $q_B=[0.60,0.20,0.20]$. The cross-entropies are $-\ln(0.80)=0.223$ and $-\ln(0.60)=0.511$ nats, respectively; Model A assigns more probability to the correct class and has lower loss. With one-hot targets, H(p)=0, so cross-entropy equals KL divergence for this example.

3.3 KL divergence is asymmetric

Let p=[0.9,0.1] and q=[0.5,0.5]. $D_{KL}(p||q)=0.9\ln(1.8)+0.1\ln(0.2)\approx 0.368$ nats. In the reverse direction, $D_{KL}(q||p)=0.5\ln(0.5/0.9)+0.5\ln(0.5/0.1)\approx 0.511$ nats. The values differ because the expectation is taken under a different distribution. $D_{KL}(p||q)$ penalises q for failing to cover probability mass where p places it; $D_{KL}(q||p)$ penalises q for placing mass in regions that p regards as unlikely. KL is non-negative but not a metric and is not symmetric.

3.4 What this means for AgriVision

- Compare train and validation cross-entropy: low training loss with high validation loss suggests overfitting or a distribution mismatch.
- A high loss despite acceptable accuracy indicates that the model is confidently wrong on some examples or gives insufficient probability to the correct class.
- Under label noise, one-hot cross-entropy can encourage memorisation of incorrect labels. Label smoothing or robust losses can reduce extreme confidence, but may also weaken learning of genuinely difficult examples.

Use natural class frequencies for the main test set. If class weights or resampling are used during training, report them because they change the effective training objective and can alter probability calibration.

Task 4 – Diagnose Numerical Computation Issues

4.1 Naive softmax overflow

Softmax is $p_i = \exp(z_i)/\Sigma_j \exp(z_j)$. For $z=[1000,999,995]$, direct exponentiation in float64 overflows because $\exp(1000)$ exceeds the representable range. The ratio is mathematically valid, but the implementation may produce infinity/infinity and NaN. For very negative logits, exponentials can underflow to zero, producing loss of precision or log(0).

Use the shift-invariant form $p_i = \exp(z_i-m)/\Sigma_j \exp(z_j-m)$, where $m=\max(z)$. For the example, shifted logits are $[0,-1,-5]$, giving exponentials [1, 0.3679, 0.0067], sum ≈1.3746, and probabilities ≈[0.7275, 0.2676, 0.0049]. Subtracting the same constant from every logit does not change softmax because the common factor $\exp(-m)$ cancels.

For loss, use log-sum-exp: $\log \Sigma_j \exp(z_j) = m + \log \Sigma_j \exp(z_j-m)$. Compute cross-entropy directly from logits with a stable library implementation rather than taking the logarithm of rounded probabilities.

4.2 Covariance condition number and optimisation

The spectral condition number $\kappa(C)=\lambda_{max}/\lambda_{min}$ for a positive-definite covariance matrix measures how elongated the feature geometry is. A large κ means some directions vary far more than others. Gradient descent then needs a small learning rate to remain stable along steep directions, while progress along shallow directions is slow; loss contours are narrow valleys. If $\lambda_{min}=0$ because features are linearly dependent, the condition number is infinite and the inverse is undefined.

Pixel features are correlated, and constant or near-constant pixels may create tiny eigenvalues. Training-set standardisation, PCA truncation, L2 regularisation and a suitable optimiser can improve conditioning. However, standardisation should not be fitted on validation/test data, and PCA can remove discriminative details, so all such choices require validation.

Task 5 – Differentiate Baseline Learning Strategies

Implement three distinct strategies on both raw features and PCA features. A fair pipeline fits each transformation on training data only and uses the same splits and scoring protocol.

Algorithm	Hypothesis space / inductive bias	Strengths on 4,096 pixels	Weaknesses and PCA effect
Softmax/logistic regression (L2)	Linear class boundaries in feature space; global additive evidence.	Fast training/inference, compact parameters, probabilities and useful baseline.	Cannot represent complex nonlinear lesion patterns without feature engineering. PCA can reduce noise/collinearity and speed training, but may discard low-variance cues.

k-nearest neighbours	Non-parametric local prediction from nearby stored examples; local constancy assumption.	Can form nonlinear boundaries without fitting a parametric function.	Distance becomes less informative in high dimensions; sensitive to scaling, background and irrelevant pixels. Prediction cost and memory grow with training set. PCA may help by denoising and reducing distance concentration.
Random forest	Ensemble of decision trees; nonlinear threshold interactions and feature subsampling.	Models nonlinear feature interactions and is relatively insensitive to monotonic scaling.	Trees on raw pixels can split on nuisance pixel thresholds and may need many nodes. PCA rotates features and can make tree splits less interpretable or less effective; measure rather than assume it helps.

Recommended implementation details: use StandardScaler + multinomial logistic regression with several C values; use standardised features and tune k and distance weighting for k-NN; tune tree count, maximum depth and minimum leaf size for random forest. For PCA pipelines, fit PCA after scaling if the scale warrants it. Record macro-F1, accuracy, training time, inference time, and serialized model size.

The three methods differ structurally, not merely by hyperparameter. Logistic regression has a low-capacity global boundary, k-NN stores examples and uses local geometry, and random forest builds a nonlinear partition. Their relative ranking is empirical and should not be asserted without the experiment.

Task 6 – Analyse Capacity, Overfitting and Underfitting

6.1 Capacity–error behaviour

Model capacity control	Low-capacity behaviour	High-capacity behaviour	What to inspect
Logistic regression: stronger vs weaker L2 (C)	Very strong regularisation can underfit: high train and validation error.	Weak regularisation permits large weights and can fit noise; validation error may rise.	Plot train and validation error against log(C); remember C is inverse regularisation strength in common libraries.
k-NN: k	Large k smooths boundaries excessively and may underfit.	k=1 can memorise training labels, producing low train error but unstable validation results.	Plot errors for several k values; test uniform and distance weighting.
Decision tree: max_depth / min_samples_leaf	Shallow trees cannot express detailed interactions.	Deep trees can memorise individual training examples and noisy labels.	Plot train/validation macro-F1 or error by depth and leaf size.

Underfitting is indicated by high training and validation error because the hypothesis class cannot capture the signal or is over-regularised. Overfitting is indicated by very low training error and a larger validation error because the model fits sample-specific detail, label noise or background correlations. The generalisation gap is validation error minus training error; it often widens at high capacity, although the precise curve is data-dependent.

6.2 Correct model selection

Choosing hyperparameters on the test set leaks information from the test outcomes into the design process. Repeatedly selecting the best test result produces an optimistically biased estimate even if each individual score is unbiased. Use a validation set or stratified k-fold cross-validation within training data for model selection, then evaluate the selected pipeline once on the held-out test set. If data is limited, nested cross-validation can separate selection from evaluation.

Required plot: for each model, graph training and validation error against its capacity parameter and mark the underfitting region, minimum-validation-error region and overfitting region. Explain each region using the model's assumptions and the observed gap, not only the shape of the curve.

Task 7 – Decompose Error into Bias and Variance

7.1 Bootstrap experiment

Use at least 10 bootstrap replicas (preferably 30–100): sample training examples with replacement, fit a low-capacity model (e.g., strongly regularised logistic regression) and a high-capacity model (e.g., deep tree or k-NN with k=1), and predict class probabilities for the same held-out examples. For a rigorous decomposition, use a defined loss such as squared error on one-hot class targets. For test point x, let $\bar{f}(x)$ be the average predicted probability vector over replicas and y*(x) the target probability vector. Then estimated bias² is the average $||\bar{f}(x)-y^*(x)||^2$, variance is the average over x of the bootstrap variability of f_b(x), and irreducible noise is the conditional target variance. For deterministic labels, an estimate of irreducible noise may be close to zero, but label ambiguity and annotator error make it nonzero.

The identity expected under the matching squared-error setup is $E[(f-y)^2] = \text{bias}^2 + \text{variance} + \text{noise}$. In a finite experiment, the equality may not hold exactly because the terms are estimated on finite samples, bootstrap replicas are correlated through the original dataset, class probabilities may be clipped, and the selected loss/averaging convention may differ. Do not claim a numerical decomposition without running the replicas.

Illustrative pattern (not measured)	Low-capacity model	High-capacity model
Bias²	Higher: simple boundary misses nonlinear patterns.	Lower on training distribution if flexible enough.
Variance	Lower: fitted model changes less across resamples.	Higher: boundary can change markedly with sampled examples.
Training error	Moderate or high.	Very low, potentially zero.
Validation error	May be stable but limited by underfitting.	Can rise due to instability and memorised noise.

7.2 MLE versus MAP for a class proportion

Suppose the probability θ of a class is estimated from n labels, with s observed positives. The Bernoulli likelihood is proportional to θ^s(1−θ)^(n−s). The maximum-likelihood estimate is θ_MLE=s/n. With a Beta(α,β) prior, the posterior is Beta(s+α,n−s+β). Its posterior mean is (s+α)/(n+α+β); the interior MAP is (s+α−1)/(n+α+β−2) when α+s>1 and β+n−s>1.

Example: s=7 positives out of n=10 gives θ_MLE=0.70. Under a Beta(2,2) prior, posterior mean=(7+2)/(10+4)=0.643, and MAP=(7+1)/(10+2)=0.667. The symmetric prior shrinks the estimate toward 0.5. At n=1,000 with 700 positives, MLE=0.700 and posterior mean=702/1004≈0.699; the prior's relative influence becomes small as data grows. A prior stabilises estimates for small samples but can introduce bias if badly chosen.

For AgriVision, Bayesian smoothing of class proportions can prevent extreme estimates in small classes. It does not automatically correct systematic label noise or sampling bias; those require explicit modelling or better annotation.

Task 8 – Analyse Gradient-Based Optimisation

8.1 Softmax regression and gradient derivation

For N examples with feature matrix X ∈ R^(N×d), one-hot targets Y ∈ R^(N×K), weights W ∈ R^(d×K) and biases b ∈ R^K, logits are $Z=XW+1b^T$. Softmax probabilities are $P_{ik}=\exp(Z_{ik})/\Sigma_j \exp(Z_{ij})$. The average cross-entropy with L2 regularisation is $L=-(1/N)\Sigma_i\Sigma_k Y_{ik} \log P_{ik} + (\lambda/2)||W||^2_F$.

For one example, $\partial L_i/\partial z_{ik} = p_{ik} - y_{ik}$. Applying the chain rule through $Z=XW$ gives $\nabla_W L = X^T(P-Y)/N + \lambda W$ and $\nabla_b L = (1/N)\Sigma_i(P_i - Y_i)$. The gradient has two parts: the prediction residual $(P-Y)$, which pushes probability toward the target class, and the feature matrix X, which distributes that error across weights. L2 adds λW, shrinking weights. A gradient-descent update is $W \leftarrow W - \eta\nabla_W L$ and $b \leftarrow b - \eta\nabla_b L$.

8.2 Optimisation strategies

Strategy	Update data	Advantages	Limitations
Batch gradient descent	All N examples per update.	Stable, low-noise direction; useful for small datasets.	One update is expensive for large N; can make slow progress per wall-clock time.
Stochastic gradient descent	One example per update.	Cheap, frequent updates; noise can help escape shallow regions.	High-variance trajectory; learning-rate scheduling is important.
Mini-batch SGD	A batch, e.g. 32 or 64, per update.	Vectorised hardware use and lower gradient variance than single-example SGD.	Batch size changes noise, memory use and number of updates per epoch.

One epoch means one pass over training examples, but the number of updates differs: batch GD uses one update per epoch, SGD approximately N updates, and mini-batch SGD approximately ceil(N/B). Compare both loss versus epoch and loss versus wall-clock time, since they answer different questions.

8.3 Learning-rate analysis

Run at least $\eta = 0.001$, 0.01 and 0.1 as starting examples, adapting them to the scaled features and observed gradient magnitude. A very small rate gives stable but slow loss reduction; a moderate rate often converges quickly; a rate that is too large may overshoot, oscillate or diverge. These exact rates are not guaranteed to suit every dataset.

Conditioning explains why a single learning rate can be poor: the largest eigenvalue of the Hessian/covariance-linked curvature constrains stability, while small-eigenvalue directions need more updates. Standardisation and PCA truncation may improve the spectrum, so compare raw and compressed inputs using the same protocol. Plot training and validation cross-entropy against epoch and wall-clock time; report divergence/instability if it occurs.

Task 9 – Organise and Examine Experimental Performance

9.1 Required result tables

Run every model at each training scale and on raw/PCA features. The following table is a fill-in template; numbers must come from the actual run.

Train images	Features	Model	Accuracy	Macro-F1	Train time (s)	Inference (ms/image)	Model size (MB)
1,000	Raw	Softmax regression	—	—	—	—	—
1,000	PCA k=chosen	Softmax regression	—	—	—	—	—
1,000	Raw	k-NN	—	—	—	—	—
1,000	PCA k=chosen	k-NN	—	—	—	—	—
1,000	Raw	Random forest	—	—	—	—	—
5,000 / 10,000 / full	Repeat all settings	All three algorithms	—	—	—	—	—

9.2 Metrics and interpretation

- Accuracy = correct predictions / all predictions. It is intuitive but can conceal class imbalance.
- Per-class precision = TP/(TP+FP); recall = TP/(TP+FN); F1 = 2·precision·recall/(precision+recall). Macro-F1 averages class F1 scores equally.
- The confusion matrix reveals directional errors. Early and Late Blight may be confused because symptoms can overlap in shape/texture; Healthy may be confused with mild disease under poor lighting. Verify these hypotheses against actual confusion-matrix cells and representative images.
- Training time generally increases with data size. k-NN stores the training set and can have costly prediction; linear models typically have compact parameters and fast inference; tree ensembles can grow with tree count and depth.
- Report the unit and measurement environment. “Inference time” should state batch size, CPU/GPU, warm-up procedure and whether preprocessing is included.

For each scale, make line plots of macro-F1 and accuracy versus number of training examples, plus bar charts for inference latency and model size. Add error bars across seeds where possible. A single test score does not establish a reliable trend.

Task 10 – Compare Theory with Experimental Behaviour

Theoretical expectation	Evidence to compare	Possible explanation if it disagrees
PCA reconstruction error falls as k rises; classification performance may plateau or decline.	Plot cumulative explained variance, reconstruction error and validation macro-F1 versus k.	Low-variance lesion cues may be discriminative; leading components may encode lighting/background.
Better conditioning can improve gradient optimisation.	Compare eigenvalue spectrum/condition number and loss-versus-time for raw and PCA inputs.	Learning rate, regularisation, solver, feature scaling or implementation may dominate the effect.
High-capacity models often show lower train error and potentially higher variance.	Capacity curves plus bootstrap bias^2/variance estimates.	Limited sample size, strong regularisation, noisy labels or a poorly chosen capacity range may shift the curve.
More training examples usually reduce estimation variance.	Compare metrics and seed variation at 1k, 5k, 10k and full scale.	Additional examples may be redundant, imbalanced, mislabeled or from a different distribution.

A strong analysis should identify both agreement and disagreement. For example, a PCA representation may explain 95% of total pixel variance but lose disease accuracy if the discarded 5% contains fine lesions. That result would not contradict PCA mathematics; it would show that variance preservation and label information are different objectives.

Task 11 – Analyse whether Deep Learning is Justified

11.1 Evidence required from the baselines

Deep learning is justified only if the tuned classical baselines leave meaningful errors that align with their structural limitations. Examine whether softmax regression underfits, whether k-NN degrades as dimensionality and dataset size increase, whether tree ensembles are too large or slow, and whether PCA improves efficiency without unacceptable class-wise loss. The final argument must quote the actual macro-F1, class recall, latency and size from Task 9.

11.2 Dimensionality, local constancy and manifold structure

- Curse of dimensionality: in 4,096 dimensions, examples become sparse and distances can concentrate. k-NN may need many more labelled images to find truly comparable neighbours, and irrelevant background pixels distort similarity.

- Local constancy: k-NN assumes nearby points tend to share labels. Lighting, rotation, camera distance and leaf pose can move images in pixel space even when the disease is unchanged.
- Smoothness: small nuisance changes should not change the diagnosis, but a pixel-wise classical model may respond to shifts and illumination. Data augmentation and convolutional inductive bias can improve robustness.
- Manifold structure: natural leaf images occupy a constrained subset of all possible pixel vectors. CNNs can learn local edges, spots, texture and larger lesion arrangements hierarchically, rather than treating each pixel as an unrelated feature.

11.3 Recommendation and trade-offs

For the three-class research setting, first retain classical baselines as transparent, inexpensive references. If the best tuned baseline remains substantially worse on held-out macro-F1 or fails on realistic lighting/device shifts, a CNN is justified because it learns spatially local, hierarchical features. A compact pretrained or mobile-oriented CNN may require more training compute and careful calibration, but can have low inference cost after quantisation. A classical linear model remains easier to inspect and may be adequate for a narrow, clean dataset.

The conclusion must be tied to evidence: state the strongest baseline, its per-class failures, how PCA changes results, and whether its memory and latency meet requirements. Do not use “deep learning is more accurate” as the sole argument.

Task 12 – Analyse the Impact of Changed Requirements

A. Impact analysis

Change	Mathematical/statistical impact	Operational implication
3 classes → 38 classes	Maximum class entropy rises from $\log_2 3 \approx 1.585$ bits to $\log_2 38 \approx 5.248$ bits under a uniform distribution. A uniform random guess has expected accuracy $1/38 \approx 2.63\%$, compared with $1/3 \approx 33.3\%$.	More output scores and more opportunities for confusions; per-class and macro metrics become essential.
Approximately 20% noisy labels	Observed targets differ from true labels; estimated training risk includes annotation error. Flexible models may memorise incorrect labels, raising variance and reducing useful signal.	Need label auditing, robust training and evaluation on a clean, representative test set.
Latency ≤50 ms and memory ≤5 MB	Compute and storage are now hard constraints rather than secondary reporting metrics.	Large forests, stored-example k-NN and uncompressed models may be unsuitable; target-device profiling is mandatory.

The entropy figure is a maximum for a uniform 38-class distribution, not a claim that the real classes are balanced. Actual entropy must be computed from observed class proportions. Label noise can either inflate apparent uncertainty or systematically bias the class distribution, depending on which labels are corrupted and how.

B. Examine the original conclusion

The original three-class model cannot be assumed to satisfy the new target: the label space has changed, and 20% noisy labels alter the training objective. Measure each candidate on the actual low-cost phone. Record model-file size and peak runtime memory separately; a 5 MB file does not guarantee total process memory under 5 MB. Report p50 and p95 single-image latency after warm-up, including required preprocessing.

Candidate	Expected sensitivity to new requirements	Evidence needed before acceptance
k-NN on raw pixels	Highly sensitive to dataset growth because examples must be stored and distances computed; adding classes increases overlap.	Memory for stored vectors, per-image latency, macro-F1 and class confusion.

PCA + regularised softmax regression	Compact and fast; more classes enlarge output weights from 3 to 38 and may need more examples to separate related classes.	Model size, phone latency, class-wise recall and calibration under noise.
Random forest	Nonlinear but tree count/depth can increase memory and latency; noisy labels can create misleading splits.	Serialized size, target-device latency and validation behaviour under corrupted training labels.
Quantized mobile CNN	Can exploit spatial structure and be small/fast, but training and quantisation require validation.	Actual int8 model size, phone p50/p95 latency, peak memory, macro-F1 and calibration.

A less noise-sensitive alternative is not guaranteed by algorithm name alone. Strong regularisation and robust training can reduce memorisation, but every candidate must be evaluated under the same controlled noise conditions.

C. Adaptation options

Option	Effect on capacity/error	Cost and risks
1. Change PCA dimension and regularisation	Lower k and stronger L2 reduce effective capacity and variance; too much compression/regularisation increases bias and may erase subtle disease cues.	Less storage and faster inference/training for linear models; choose k and λ by validation macro-F1, not variance alone.
2. Noise-robust training	Use label smoothing, generalized cross-entropy or symmetric cross-entropy; combine with suspected-label review or confident-example selection.	May reduce overconfidence and memorisation of wrong labels. Excessive smoothing/robustness can underfit clean hard examples; validate on a clean set.
3. Compress/simplify model	Quantise weights/activations to int8, prune where supported, or distil a larger teacher into a compact student.	Reduces model size and often latency, but quantisation/pruning can reduce recall for visually subtle classes; measure on-device.
4. Use a mobile-oriented CNN	Convolutions exploit spatial locality and hierarchical features; transfer learning can improve data efficiency.	Potentially better visual representation, but more engineering/training complexity. Quantisation and runtime operator support determine real performance.

Label smoothing example: replace a one-hot target with $1-\varepsilon$ for the true class and $\varepsilon/(K-1)$ for each other class. It prevents a model from being trained toward absolute certainty, but it does not identify which labels are wrong. Under 38 classes, choose ε conservatively and validate calibration and macro-F1.

D. Evidence-based conclusion

Decision: MODIFY the original plan and allow a REDESIGN if measured deployment tests fail. Keep PCA + regularised softmax as the compact classical baseline because it is fast, interpretable and comparatively easy to fit within 5 MB. However, do not retain it as the final model merely because it worked on three clean classes. The 38-class label space, approximately 20% noisy labels and smartphone constraints change both the statistical problem and the deployment objective.

The final model-selection rule should be: (1) train with a documented noise-aware protocol; (2) compare a tuned PCA-linear baseline against a compact quantized CNN; (3) evaluate on a clean, group-separated test set containing all 38 classes; (4) require macro-F1 and per-class recall to meet the application's predefined threshold; and (5) accept only a candidate whose measured p95 latency is below 50 ms and model size is below 5 MB on the target phone. If the linear baseline meets all quality thresholds, retain it for simplicity. If it misses important classes or is not robust to nuisance variation, redesign around the compact CNN, provided its on-device measurements meet the limits.

This recommendation follows the mathematics: 38 classes increase the output entropy ceiling; noisy labels make the observed risk differ from true-label risk and increase the danger of high-capacity memorisation; PCA can improve conditioning but may discard low-variance disease details; and model compression trades capacity against runtime. The

actual choice remains conditional on measured results. Since no dataset run or phone measurements were supplied, no claim is made here that either candidate has already met 50 ms or 5 MB.